

Orange Money Bank2Wallet

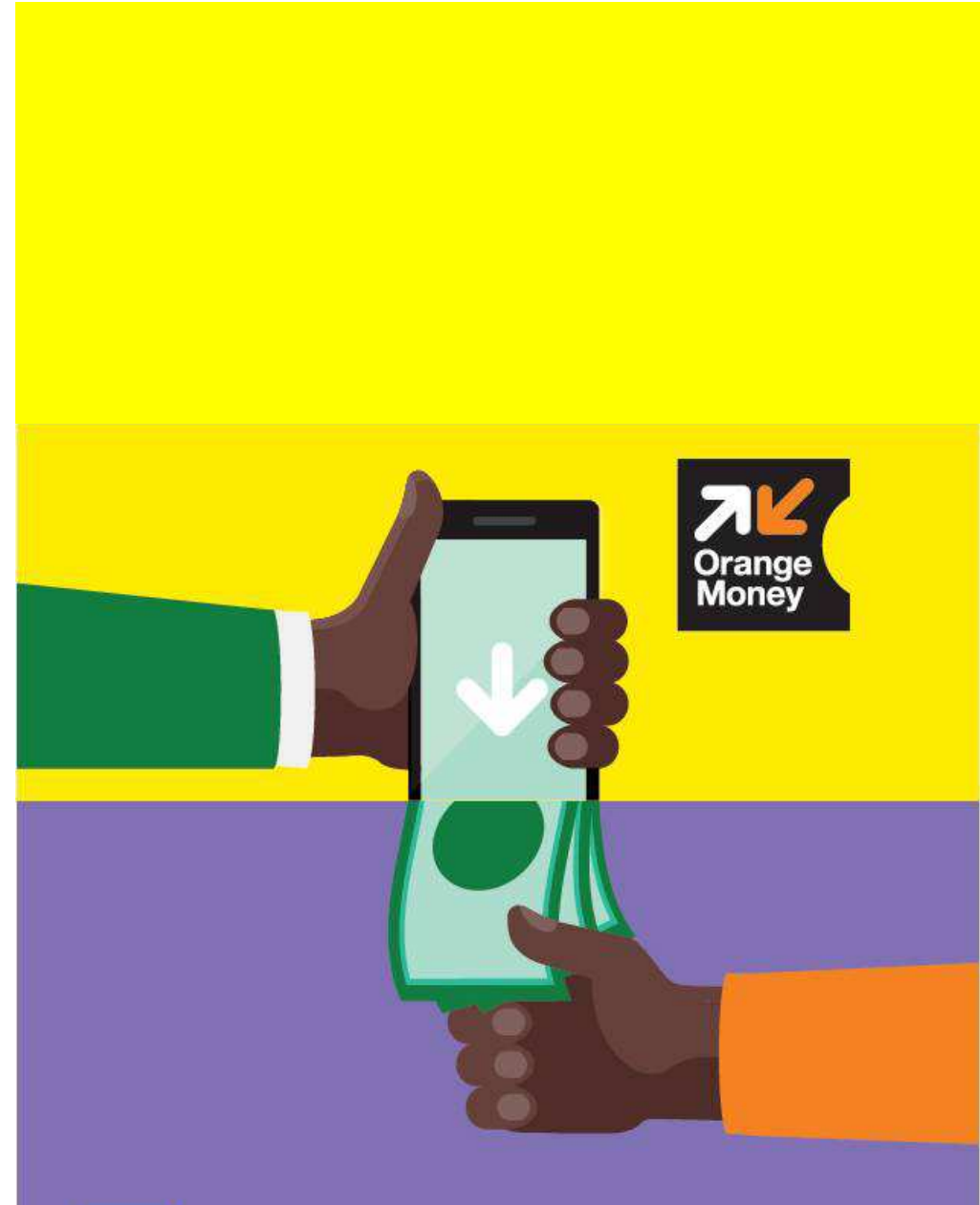
Information:

Management of transfer errors
on B2W interface

Renaud MICHEL - TGI/OLS/CCMB
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Internal Orange



Part 1 : Transfer-status-inquiry



Specification of the BankAPI says:

6 TransferStatusInquiry: transfer's status check

to check the status of a transfer.

in response :

- CBAReferenceNo filled <=> original transaction OK on bank side
- CBAReferenceNo empty <=> original transaction not found or not valid on bank side (i.e. cancellation has to be done on mobile operator side)

Operation	TransferStatusInquiry			
Request (TransferStatusInquiry) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	OR150313C9OL8X
requestToken	string	128	Not used (empty field)	empty
requestType	string	20	TRANINQ (Transaction status Inquiry)	TRANINQ
affiliateCode	string	3	country affiliate code of the bank.	EML
externalRefNo	string	16	reference (requestId) of original transaction which has to be checked	OR151023OSR023
Response (TransferStatusInquiryResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR150313C9OL8X
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	OR150313C9OL8X
CBAReferenceNo	string	16	original transaction reference if found on bank side, empty if transaction not found	FKGI22541

Usage principle:

The “t-s-i” is a second chance to execute the transfer, in case of fail of the initial request.

The process and response to a “t-s-i” is similar to the initial transfer.

Implementation :

- Always reply the “t-s-i” with a responseCode + responseMessage
- Response codes should indicate the result of the processing of the “t-s-i” (‘000 -Success’ = bank has processed the “t-s-i”)
- Only in case of “000-Success” of the “t-s-i”, the CBAReferenceNo has to indicates the result of the inquiry in the core banking :
 - >If the transaction was executed by the core banking, fill it with the bank internal reference (ex: “FKIxxx541”)
 - >If the transaction wasn't found or failed for any reason in the core banking, leave it blank
- Beware of ID confusion: inquiry should focus on the processing made by the bank after Orange transaction “externalRefNo” (=“ORxxx023” in the sample, which the requestId of the transfer made before). Not to be confused with the “requestId” (=“ORxxxL8X”), which is the ID of the “t-s-i” request.

Transfer indirect success (i.e. success after a “transaction-status-inquiry”)

Détails de la transaction Orange Money :

Emetteur	Receveur	Date	OM Transaction Id	Type	Currency	Montant	Statut	Message d'erreur
7702100041		2020-06-08 16:28:36	ES200608.1828.A66391	get-balance	Ouv		200	Success
main	7702100041	2020-06-08 16:28:36	7702920041201906081828	authorization-CI	Ouv	12500.00 Ouv	200	Success
main	7702100041	2020-06-08 16:31:37	CI200608.1831.B00517	cashin-credit	Ouv	12500.00 Ouv	200	Success
7702100041		2020-06-08 16:31:38	ES200608.1831.A66396	get-balance	Ouv		200	Success
7702100041		2020-06-08 16:31:38	GF200608.1831.B00167	get-fees	Ouv	12500.00 Ouv	200	Success

Nombre total de résultats :5 enregistrement(s)

Détails de la transaction Banque :

Date Transaction	Type	Alias	Currency	Montant	Code requête	Code réponse	Message réponse	Référence n°
2020-06-08 16:28:36	bank2wallet	BNKSXQKP657895124567	Ouv	12500.00 Ouv	OR200608ATLQ32		SoapFault: Error Fetching http headers	
2020-06-08 16:31:37	transaction-status-inquiry	BNKSXQKP657895124567	Ouv	12500.00 Ouv	OR200608U38APS	000	Success	OMO20w265071

Nombre total de résultats :2 enregistrement(s)

Situation

The B2W initial request is not responded due to applicative error :”soap fault” (may happen after network connection issue with the Bank).

A “t-s-i” request is send by Orange 2 minutes latter, which is replied by the Bank “000-Success” and the Bank reference number of the transfer (CBAReferenceNo), confirming that the transfer was executed.

This wallet of the customer is credited (“cashin-credit”)

Transfer indirect error (i.e. error after a “transaction status-inquiry”)

Transaction MBRCH.20200614212724.706, Type bank2wallet, Status: 60160								
Orange Money Transaction details:								
Payer	Payee	Date	OM Transaction Id	Type	Currency	Amount	Status	Error Message
621409481		2020-06-14 21:27:24	ES200614.2127.B73414	get-balance	GNF		200	Success
main	621409481	2020-06-14 21:27:24	629335428202006142127A	authorization-CI	GNF	200000.00 GNF	200	Success
Total results number:2 record(s)								
Bank Transaction details:								
Unsubscription Date	Type	Alias	Currency	Amount	Request code	Response Code	Response Message	Reference Number
2020-06-14 21:27:24	bank2wallet	ORBKGNGN080486500110	GNF	200000.00 GNF	OR200614G645WP		SoapFault: Error Fetching http headers	
2020-06-14 21:28:34	transaction-status-inquiry	ORBKGNGN080486500110	GNF	200000.00 GNF	OR200614TCLVM6	E11	Operation in Timeout. Use TransferStatusInquiry to check status	
2020-06-14 21:29:13	transaction-status-inquiry	ORBKGNGN080486500110	GNF	200000.00 GNF	OR200614UGP0RL	E11	Operation in Timeout. Use TransferStatusInquiry to check status	
2020-06-14 21:29:53	transaction-status-inquiry	ORBKGNGN080486500110	GNF	200000.00 GNF	OR200614F45WE1	E16	Insufficient Balance	
Total results number:4 record(s)								

Remark :
In case of a W2B transfer, this error pattern results in the automatic refund of the wallet

Situation:

The B2W initial request is failing on Bank side due to an applicative error “Soap fault”.

The 2 first “t-s-i” are all failing on Bank side due to a time out (of the core banking).

On the last “t-s-i”, Bank is finally responding an error “E16-Insufficient Balance” with an empty reference number (CBAResponseNo), confirming that the core banking hasn’t executed the transfer..

No more action is performed by Orange, W2B transfer is considered not executed by the Bank and the customer is noticed that his Bank account balance is too low to sustain the transfer.

Part 2: Cancel-transfer



Specification of BankAPI says:

7 CancelTransfer: transfer cancellation

to cancel a AccountToWalletTransfer or WalletToAccountTransfer operation

Operation	CancelTransfer			
request (CancelTransfer) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	OR151023ITGA5Y
requestToken	string	128	Not used (empty field)	empty
requestType	string	20	CANCELTRAN (Transaction Cancelation request)	CANCELTRAN
affiliateCode	string	3	country affiliate code of the bank.	EML
externalRefNo	string	16	banking reference (CBAResponseNo) of original transaction which has to be cancelled	FKGI22541
Response (CancelTransferResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR151023ITGA5Y
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	OR151023ITGA5Y
CBAResponseNo	string	16	banking reference of this cancellation	FKGI25894

Usage principle:

The “cancel-transfer” provides a way for Orange to notice the bank to cancel a transfer if a fatal error happened **AFTER the Bank confirmed that the execution of the transfer** (ex: Tango configuration issue).

Cancel-transfer : success

Détails de la transaction Orange Money :

Emetteur	Receveur	Date	OM Transaction Id	Type	Currency	Montant	Statut	Message d'erreur
7702400024		2020-06-04 01:31:28	ES200604.0331.B57038	get-balance	OUV		200	Success
main	7702400024	2020-06-04 01:31:32	PP200604.0331.A00104	c2c-credit	OUV	10.00 OUV	00084	No Service Charge is defined

Nombre total de résultats :2 enregistrement(s)

Détails de la transaction Banque :

Date Transaction	Type	Alias	Currency	Montant	Code requête	Code réponse	Message réponse	Référence n°
2020-06-04 01:31:28	bank2wallet	BNKSBZXY0011110394667	OUV	10.00 OUV	OR200604U3FJOB	000	Success	20200213715
2020-06-04 01:31:32	cancel-b2w	BNKSBZXY0011110394667	OUV	10.00 OUV	OR20060489L40P	000	Success	20200213717

Nombre total de résultats :2 enregistrement(s)

SMS envoyé

Situation:

The B2W initial request is confirmed by the Bank with a “000-Success” and the appropriate reference number (CBARreferenceNo) But a fatal error is raised when trying to credit the wallet of the customer : tango raises a “No service change is defined” that prevents to credit the wallet of the customer

Given that the Bank has already confirmed the b2w-transfer (i.e. debited the customer bank account), Orange sends a “cancel-b2w” to notice the bank that the transfer should be cancelled.

The Bank replies the cancel with a “000-Success” with another reference number (CBARreferenceNo) which is representing the re credit of the customer account.

Cancel-transfer : error

Params

Transaction MBRCH.20200616183055.642, Type bank2wallet, Status: 630

Orange Money Transaction details:

Payer	Payee	Date	OM Transaction Id	Type	Currency	Amount	Status	Error Message
90922876		2020-06-16 19:30:55	ES200616.1930.B00559	get-balance	FCFA		200	Success
main	90922876	2020-06-16 19:31:14	PP200616.1931.A09289	c2c-credit	FCFA	11000000.00 FCFA	00410	The transaction amount is more than the maximum value defined for this service

Total results number:2 record(s)

Bank Transaction details:

Unsubscription Date	Type	Alias	Currency	Amount	Request code	Response Code	Response Message	Reference Number
2020-06-16 19:30:55	bank2wallet	ORBKNENI7226690400100	FCFA	11000000.00 FCFA	OR2006160W675R	000	Success	OM972133964
2020-06-16 19:31:14	cancel-b2w	ORBKNENI7226690400100	FCFA	11000000.00 FCFA	OR200616SNJ9AG	E01	Bank Technical Error	

Total results number:2 record(s)

Situation:

The B2W initial request is confirmed by the Bank with a “000-Success” and the appropriate reference number (CBARreferenceNo) But a fatal error is raised when trying to credit the wallet of the customer is raised by Tango that prevents to credit the wallet of the customer.

Given that the Bank has already confirmed the b2w-transfer (i.e. debited the customer bank account), Orange sends a “cancel-b2w” to notice the bank that the transfer should be cancelled.

The Bank replies the cancel with a “E01-Technical Error” without new reference number (CBARreferenceNo) which confirms that the Bank couldn’t re credit of the customer account.

Merci

Thanks

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